

Designation: F 4123-24

Standard Specification for Shape Memory Alloy Actuator Wire

This standard was issued on June 1, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers nickel-titanium (NiTi) shape memory alloy wire for actuator and medical device applications

1.2 The shape memory alloy wire covered by this specification are intended for use in robotic actuators, aerospace mechanisms, biomedical devices, and smart structures requiring thermally-activated shape change.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

F 2063 Standard Specification for Wrought Nickel-Titanium Shape Memory Alloys for Medical Devices

F 2516 Test Method for Tension Testing of Nickel-Titanium Superelastic Materials

E 1876 Test Method for Dynamic Young's Modulus, Shear Modulus, and Poisson's Ratio by Impulse Excitation

E 1269 Test Method for Determining Specific Heat Capacity by Differential Scanning Calorimetry

3. Terminology

3.1 Definitions:

3.1.1 shape memory effect—the ability of certain alloys to recover a predetermined shape upon heating

3.1.2 transformation temperature—the temperature at which the alloy transforms between austenite and martensite phases

3.1.3 recovery stress—the stress generated during constrained shape recovery

4. Classification

4.1 shape memory alloy wire shall be classified according to transformation temperature range and wire diameter.

4.1.1 **Type LT**—Low temperature, $A_f = 40-60^{\circ}\text{C}$, diameter 0.1-2.0 mm

4.1.2 **Type MT**—Medium temperature, $A_f = 70-90^{\circ}\text{C}$, diameter 0.1-2.5 mm

4.1.3 **Type HT**—High temperature, Af = 100-120°C, diameter 0.1-3.0 mm

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

5.1.1 ASTM designation and year of issue

5.1.2 Type (LT, MT, or HT)

5.1.3 Wire diameter (mm)

5.1.4 Transformation temperatures (As, Af, Ms, Mf if critical)

5.1.5 Length (meters)

5.1.6 Quantity

5.1.7 Heat treatment condition (trained or as-drawn)

6. Materials and Manufacture

6.1 Wire shall be produced from high-purity nickel-titanium alloy by vacuum induction melting and hot working.

6.2 Final wire diameter shall be achieved through cold drawing with intermediate annealing.

6.3 Shape memory training shall be performed if specified, using appropriate thermal-mechanical cycling.

7. Chemical Composition

7.1 The material shall conform to the chemical composition requirements specified in Table 1.

Element	Composition, %
Nickel (Ni)	55.0-56.5 wt%
Titanium (Ti)	Balance
Carbon (C), max	0.05%
Oxygen (O), max	0.05%
Nitrogen (N), max	0.005%
Hydrogen (H), max	0.005%

8. Mechanical and Physical Requirements

Property	Requirement	Test Method
Transformation temp Af	Within $\pm 5^{\circ}\text{C}$ of target	DSC per E 1269
Recovery strain	$\geq 6\%$	Bend and free recovery
Recovery stress	$\geq 400\text{ MPa}$	Constrained recovery
Fatigue life	$\geq 10,000$ cycles at 4% strain	Cyclic testing

9. Dimensions and Permissible Variations

9.1 Wire diameter tolerance shall be ± 0.010 mm for diameters < 0.5 mm.

9.2 For diameters ≥ 0.5 mm, tolerance shall be $\pm 2\%$ of nominal diameter.

9.3 Surface finish shall be bright and smooth without deep scratches or pits.

9.4 Straightness shall be ≤ 5 mm deviation per meter length.

10. Workmanship, Finish, and Appearance

10.1 Wire shall be free from cracks, laps, inclusions, and surface defects that could serve as crack initiation sites during cyclic loading.

11. Sampling

11.1 Transformation temperatures shall be measured for each production batch. Mechanical properties verified on representative samples.

12. Test Methods

12.1 **Transformation Temperature**—Determine A_s , A_f , M_s , M_f by differential scanning calorimetry (DSC) per ASTM E 1269.

12.2 **Shape Memory Testing**—Deform wire in martensite condition, then heat above A_f and measure recovery strain and stress.

12.3 **Fatigue Testing**—Cycle wire through transformation by thermal or mechanical means and monitor degradation of properties.

13. Inspection

13.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

14. Rejection and Rehearing

14.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

15. Certification

15.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

16. Product Marking

16.1 Wire spools shall be labeled with alloy composition, wire diameter, transformation temperatures, lot number, and production date.

17. Packaging and Package Marking

17.1 Wire shall be wound on plastic spools and packaged in sealed bags with desiccant. Storage at room temperature in dry conditions.

18. Keywords

18.1 shape memory alloy; NiTi; nitinol; actuator; smart material; superelastic; transformation